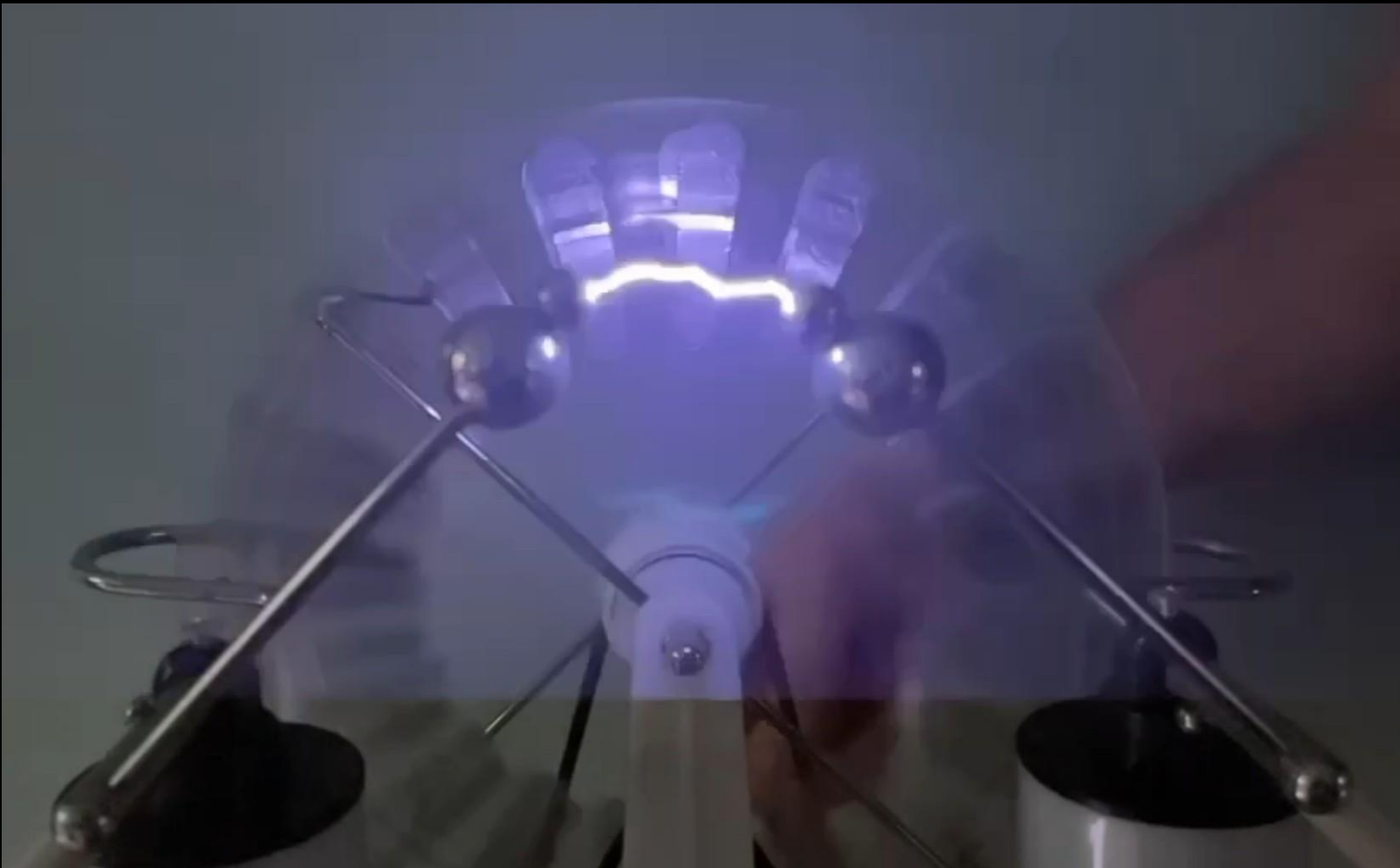




WIMSHURST ELECTROSTATIC GENERATOR

Investigatory Project: From Electrostatic Force to High Voltage, Power & Electromagnetic Emissions

Presentation follows the 60-entry research logbook progression

The Investigation Follows the Logbook

Each phase answers a question raised by the phase before it.

1 Concept & Background

2 Electrostatic Theory

3 Styrofoam Force Demo

4 Experimental Planning

5 Spark & Leyden Jars

6 Voltage, Current & Power

7 EM Emission

8 Applications & Testatika

Final Analysis & Presentation

PHASE 1

Conceptualization & Background Research

Entries 1–8

Guiding Question

What does a Wimshurst machine do, and what can we investigate with it?

WIMSHURST ELECTROSTATIC GENERATOR INVESTIGATORY PROJECT REPORT AND DEVELOPMENT LOGBOOK

Project Purpose: To investigate the operation of the Wimshurst electrostatic generator, analyze the production of high voltage and low current, understand the role of capacitance using Leyden jars, measure voltage by spark gap, evaluate power output, detect electromagnetic emission of the spark, and explore practical applications including the Tesla coil electrostatic power system.

INVESTIGATORY PROJECT DEVELOPMENT LOGBOOK – 60 ENTRIES

PHASE 1 Conceptualization & Background Research (Entries 1–7)

PHASE 2 Electrostatic Theory & Basic Properties (Entries 8–15)

PHASE 3 Experimental Planning & Initial Measurements (Entries 16–24)

PHASE 4 Spark & Leyden Jar Investigation (Entries 25–35)

PHASE 5 Voltage, Power & Energy Evaluation (Entries 36–42)

PHASE 6 EM Emission Detection (Entries 43–48)

PHASE 7 Practical Application & Extended Example (Entries 49–55)

PHASE 8 Analysis, Conclusions & Next Steps

LEGEND (How to read this logbook)

Photo Placeholder - Replace with your actual experimental photograph

Observation / Data Recording

Question / Hypothesis

Conclusion / Finding

Figure 1. The Wimshurst Electrostatic Generator (Typical Laboratory Model)

Figure 2. Charge Density, Electric Field, and Air Breakdown

Figure 3. Measuring the Spark Gap

Figure 4. Effect of Leyden Jars on Discharge

Figure 5. Typical Electrical Characteristics (Example)

Figure 6. Air Break Detection Setup

Figure 7. Tesla Coil Electrostatic Power System (Example)

Figure 8. From Electrostatics to Useful Power

Key Finding: The Wimshurst machine generates a high-voltage, low-current electrostatic discharge (ESD) that can be used to power a Tesla coil, which in turn can power a radio receiver.

What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

What Does a Wimshurst Machine Do?

Mechanical motion separates charge and creates a very large potential difference.

- Two insulating disks rotate in opposite directions.
- Metal sectors, neutralizer brushes and collectors reinforce charge separation.
- Opposite charges accumulate at the output terminals.
- A sufficiently strong electric field breaks down the air and produces a spark.

Mechanical energy → charge separation → high voltage → spark

WIMSHURST ELECTROSTATIC GENERATOR
INVESTIGATORY PROJECT REPORT AND DEVELOPMENT LOGBOOK

Project Purpose
To investigate the operation of the Wimshurst electrostatic generator, analyze the production of high voltage and low current, understand the role of capacitance using Leyden jars, measure voltage by spark gap, evaluate power output, detect electromagnetic emissions of the spark, and explore practical applications including the Tesla coil electrostatic power system.

LEGEND (How to read this logbook)

- Photo Placeholder – Replace with your actual experimental photograph
- Observation / Data Recording
- Question / Hypothesis
- Conclusion / Finding

INVESTIGATORY PROJECT DEVELOPMENT LOGBOOK – 60 ENTRIES

PHASE 1: Conceptualization & Background Research (Entries 1–7)

1. Selected the Wimshurst machine as the subject after discussions on electrostatic generators and high voltage.
2. Researched James Wimshurst (1832–1902) and the history of electrostatic influence machines in the late 1800s.
3. Identified main investigation goals: high voltage, charge accumulation, capacitance, voltage measurement, power, and practical applications.
4. Defined key research questions (initial list and scope).
5. Reviewed basic safety requirements for high-voltage electrostatic experiments.
6. Surveyed available Wimshurst machines and selected the one to be used.
7. Studied influence machine designs: Wimshurst, Holtz, Pidgeon, and Pidgeon (1898) used variable capacitance via sector wiring.

PHASE 2: Electrostatic Theory & Basic Properties (Entries 8–15)

8. Reviewed electric charge: positive, negative, conductors, insulators, and charge conservation.
9. Studied surface charge density ($\sigma = Q/A$) and its effect on electric field strength.
10. Reviewed electric field (E) and its relationship to voltage: $E = V/d$ for parallel plates.
11. Reviewed capacitance (C) and its core relationship: $Q = C \cdot V$.
12. Learned about Leyden jars (capacitors): storing charge and increasing energy.
13. Derived energy stored in a capacitor: $E = \frac{1}{2} C V^2$.
14. Studied air breakdown and dielectric strength of air ($\sim 3 \text{ kV/mm}$ under normal conditions).
15. Read about spark phenomena and the conditions required for electrical breakdowns in air.

Figure 1. The Wimshurst Electrostatic Generator (Typical Laboratory Model)

Figure 2. Charge Density, Electric Field, and Air Breakdown

Surface Charge Density: $\sigma = Q/A \text{ (C/m}^2\text{)}$
Higher σ on the electrode surface produces a stronger electric field.

Electric Field Between Plates: $E = V/d \text{ (N/C)}$
Electric field increases with voltage (V) and decreases with distance (d).

Air Breakdown: Breakdown occurs when E reaches the dielectric strength of air ($\sim 3 \text{ MV/m}$).
When E exceeds this value, a spark forms and air becomes conductive.

Coulomb's Law: The electrostatic force between two point charges is $F = k \frac{q_1 q_2}{r^2}$
where $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.
Like charges repel (F outward).
Unlike charges attract (F inward).

Figure 2A. Styrofoam Ball Attraction and Repulsion Experiment

1. Neutral ball attracted: A neutral Styrofoam ball is attracted to the charged terminal due to charge polarization.
2. Contact and charge transfer: When it touches, it acquires the same (induced) charge.
3. Repelled from same terminal: The now positively charged ball is repelled.
4. Attracted to opposite terminal: It is attracted to the negative terminal. After contact, it becomes negative and is repelled again. The cycle repeats.

Key Finding: The Wimshurst machine can move a lightweight ball even though the continuous current is extremely small. Electrostatic forces depend on charge and electric field, not on a large current. This demonstrates the power of high voltage and charge accumulation.

PHASE 3: Experimental Planning & Initial Measurements (Entries 16–24)

16. Planned experiments to observe sparks with and without Leyden jars.
17. Designed to measure maximum repeatable spark gap (distance) as a method to estimate voltage.
18. Prepared safety protocols for gaps, conditions, and estimated voltage.
19. Identified variables: machine speed, humidity, presence of Leyden jars, gap distance.
20. Selected measuring tools: ruler, voltmeter, hygrometer, stopwatch, voltmeter.
21. Prepared battery-powered AM radio for EM emission test.
22. Outlined safety procedures: insulated handle, one-hand rule, no touching terminals, discharge before adjustments.
23. Conducted pre-test inspection of all components.
24. Started the machine and recorded baseline observations.

PHASE 4: Spark & Leyden Jar Investigation (Entries 25–35)

25. Connected Leyden jars to the machine.
26. Observed a single, bright, energetic spark across the gap.
27. Measured maximum repeatable gap (with jars).
28. Listed to the usual "snap crack" at each discharge.
29. Disconnected the Leyden jars.
30. Observed a dim, rapid blue discharge across the gap.
31. Measured maximum repeatable gap (without jars).
32. Compared discharge frequency: faster without jars.
33. Compared brightness and appearance of discharges.
34. Compared energy of discharges (with vs. without jars).
35. Formulated the question: Why do Leyden jars cause much stronger, further-reaching sparks?

Figure 3. Measuring the Spark Gap

The maximum gap (d) at which a spark will just jump consistently is recorded. This distance is used to estimate the voltage.

Figure 3A. Example Spark Measurement

Measured maximum gap (d): _____ mm
Condition notes: _____
Estimated voltage $V = E \cdot d$
(Use $E \approx 3 \text{ kV/mm}$ as a first approximation.)

Figure 4. Effect of Leyden Jars on Discharge

With Leyden Jars Connected: Slower discharges, No visible current, Greater stored energy per discharge, Larger maximum gap (further voltage).

Without Leyden Jars: Rapid, dim discharges, No visible current, Much less energy per discharge, Smaller maximum gap (lower voltage).

Key Finding: Leyden jars increase capacitance, allowing much more charge to be stored and discharged, producing higher voltage and more energetic sparks.

Figure 5. Typical Electrical Characteristics (Example)

Estimated Voltage both Leyden jars: $\sim 25\text{--}30 \text{ kV}$
Stored Energy per Discharge: $\sim 0.2\text{--}0.3 \text{ J}$
Charge Stored (Q): $\sim 1\text{--}2 \mu\text{C}$
Peak Current during Spark: $\sim 0.1\text{--}1 \text{ A}$ (for μs)
Peak Power $P = V \cdot I$: $\sim 10 \text{ W} - 100 \text{ W}$ (Wp Value)
Note: Values vary with humidity, jar capacitance, humidity, gap distance, and measurement method.

Figure 5A. Conceptual Power Flow

Charge Stored (Capacitance) → Discharge (High Voltage) → Resistance (Power) → Spark (Light, Sound, Heat)

Figure 6. AM Radio Detection Setup

With Leyden Jars: Receives stations to be heard, pulses from the spark discharge.

Figure 6A. Radio Signal Comparison

With Leyden Jars: Stronger, louder clicks.

Without Leyden Jars: Weaker, faster, softer clicks.

Figure 7. Testable Electrostatic Power System (Example)

Resistor (e.g., $10 \text{ k}\Omega$)

Figure 7A. Power Characteristics to Stimulate Power

The spark from the Wimshurst machine produces broadband electromagnetic pulses easily detected by a simple AM radio.

Next Finding: Use a high-voltage probe and oscilloscope.

PHASE 5: Voltage, Power & Energy Evaluation (Entries 36–43)

36. Calculated voltage using $V = E \cdot d$.
37. Estimated energy stored in Leyden jars: $E = \frac{1}{2} C V^2$.
38. Determined typical capacitance C of Leyden jars.
39. Estimated current during discharge: $I = Q/dt$.
40. Estimated current during discharge: $I = Q/dt$.
41. Compared discharge frequency (slow vs. fast) using stopwatch or high-speed observation (if available).
42. Compared energy of discharges (with vs. without jars).
43. Concluded: High voltage, extremely low current, high instantaneous power in a very short pulse.

PHASE 6: EM Emission Detection (Entries 44–48)

44. Set up battery-powered AM radio near the machine.
45. Tuned to the static/fuzzing region between stations.
46. Observed strong clicking/teasing, synchronized with sparks.
47. Compared AM radio response with and without Leyden jars.
48. Concluded the spark emits electromagnetic pulses.

PHASE 7: Practical Application & Historical Example (Entries 49–55)

49. Researched electrostatic power systems.
50. Studied Tesla coil: electrostatic power system (1930s).
51. Examined the electrostatic motor in Tesla's.
52. Examined the electrostatic motor in Tesla's.
53. Compared strengths and limitations with DC systems.
54. Considered advantages for remote-control applications.
55. Reflected on modern relevance of electrostatic systems.

PHASE 8: Analysis, Conclusions & Next Steps

56. Reviewed all data and observations.
57. Answered research questions from initial list.
58. Summarized key findings and conclusions.

Overall Conclusions

The Wimshurst machine generates very high voltages with extremely low current.

Next Steps

Use a high-voltage probe and oscilloscope.

PHASE 2

Electrostatic Theory Development

Entries 9–17

Guiding Question

Which quantities explain charge buildup, electric field, capacitance and air breakdown?

What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

WIMSHURST ELECTROSTATIC GENERATOR

Investigatory Project Report and Development Logbook

Project Purpose
To investigate the operation of the Wimshurst electrostatic generator, analyze the production of high voltage and low current, understand the role of capacitance using Leyden jars, measure voltage by spark gap, evaluate power output, detect electromagnetic emissions of the spark, and explore practical applications including the Tesla coil electrostatic power system.

INVESTIGATORY PROJECT DEVELOPMENT LOGBOOK – 60 ENTRIES

PHASE 1
Conceptualization & Background Research
(Entries 1–7)

1. Selected the Wimshurst machine as the subject of the project.
2. Researched James Wimshurst (1832–1903) and electrostatic machines in the late 1800s.
3. Identified main investigation goals: high-voltage, charge accumulation, capacitance, voltage measurement, power, and practical applications.
4. Defined key research questions and scope.
5. Reviewed basic safety requirements for high-voltage electrostatic equipment.
6. Surveyed available Wimshurst machines and selected the one used in experiments.
7. Studied influence machine designs: Wimshurst, Holtz, Folger, and others. Polymers used variable capacitance vs. metal wiring.

PHASE 2
Electrostatic Theory & Fundamentals
(Entries 8–15)

PHASE 3
Electrostatic Force Demonstration – Styrofoam Ball Attraction & Repulsion
(Entries 16–22)

PHASE 4
Spark Breakdown & Spark-Gap Measurement
(Entries 23–30)

PHASE 5
Leyden Jars – Capacitance & Energy
(Entries 31–38)

PHASE 6
Voltage, Current & Power
(Entries 39–45)

PHASE 7
Electromagnetic Emission & AM Radio Detection
(Entries 46–53)

PHASE 8
Applications – Tesla Coil Electrostatic Power System
(Entries 54–60)

LEGEND

- Photo Placeholder – replace with actual experimental photograph
- Observation / Data Recording
- Question / Hypothesis
- Conclusion / Finding

PHOTO PLACEHOLDER
Replace with your actual photograph of the Wimshurst machine used in your experiment.

The Wimshurst Electrostatic Generator (Typical Laboratory Model)

Charge Density & Electric Field

Higher charge density (σ) on electric surfaces → stronger electric field (E) → when E exceeds the breakdown strength → spark (air becomes ionized).

$E = V/d$

$E_{\text{break}} \approx 3 \text{ kV/mm}$

Voltage needed to jump a gap (d):
 $V_{\text{break}} \approx E_{\text{break}} \times d$

Air Breakdown (Spark Condition)

Breakdown occurs when the electric field reaches the dielectric strength of air.

Capacitance & Energy

$Q = C \cdot V$

$E = \frac{1}{2} C V^2$

Leyden jars increase C → store more charge (Q) → store more energy (E).

Styrofoam Ball Interaction Sequence

1. Neutral ball approaches charged terminal.
2. Ball touches positive terminal and acquires positive charge.
3. Ball is repelled from positive terminal.
4. Ball attracted toward negative terminal, touches and may charge change.

Measuring the Spark Gap

The maximum gap at which the spark will just jump consistently is recorded. Distance (d) used to estimate the voltage.

Your Wimshurst Spark Photo

Replace with your actual photograph of the spark gap using your ruler or caliper.

Typical Spark Appearance (Wimshurst Spark)

Replace with your actual photograph of a strong spark with Leyden jars connected.

Effect of Leyden Jars

With Leyden Jars Connected

- Larger discharges (less frequent)
- Thick, bright, powerful spark
- Greater stored energy per discharge
- Larger maximum spark gap (higher voltage)

Without Leyden Jars

- Rapid discharge (less frequent)
- Thin, faint blue discharge
- Much less energy per discharge
- Smaller maximum spark gap (lower voltage)

Voltage Estimate Example

$d = 12 \text{ mm (0.012 m)}$
 $V = E_{\text{break}} \times d$
 $= 3,000 \text{ V/mm} \times 12 \text{ mm}$
 $\approx 36,000 \text{ V (36 kV)}$

Typical Current Measurement

$I = 5 - 20 \text{ }\mu\text{A (average)}$
(Example value)
 $I_{\text{avg}} \approx 0.000005 - 0.000020 \text{ A}$

Power Estimate Example

$P = V \times I$
 $= 36,000 \text{ V} \times 10 \text{ }\mu\text{A}$
 $= 36,000 \text{ V} \times 0.000010 \text{ A}$
 $= 0.36 \text{ W (instantaneous)}$
Low power!

AM Radio Detection Setup (Battery-Powered Radio)

Place the radio near the spark gap for clarity in recording.

Radio Signal Comparison

Without Leyden Jars

- Rapid spark skips
- Lower amplitude
- Weaker signal

With Leyden Jars

- Slower, longer sparks
- Higher amplitude
- Stronger signal

Electrostatic Power: From Generation to Useful Power

Wimshurst Electrostatic Power Plant (1900s)

Historical photograph – replace if found.

Hydroelectric Electricity Plant is a generator that converts the kinetic energy of flowing water into electrical energy. When connected, this portable electrostatic generator → capacitors (Leyden jars).

Wimshurst Investigatory Project • Phase 2

05

Four Quantities to Keep Separate

High voltage, charge, capacitance and current describe different things.

Charge (Q)

How much electrical charge has accumulated.

Voltage (V)

Electrical potential difference between the terminals.

Capacitance (C)

How much charge can be stored for a given voltage.

Current (I)

Rate at which charge flows.

$$Q = C V \quad V = Q / C \quad \sigma = Q / A$$

Key idea

Low effective capacitance means only a small amount of charge is needed to reach a very high voltage. Surface charge density helps determine the electric field at the electrodes.

Air Breakdown and Stored Energy

The spark appears when the electric field becomes strong enough to ionize air.

$$E \approx V / d$$

Electric field

Higher voltage and smaller separation generally increase electric-field strength. Electrode curvature also concentrates the field.

$$\text{Air breakdown} \approx 3 \text{ kV/mm}$$

Air becomes conductive

When the field reaches the local dielectric strength of air, ionization creates a conductive path and a spark can form.

$$E_{\text{stored}} = \frac{1}{2} C V^2$$

Leyden jars store energy

Increasing capacitance lets more charge and more energy accumulate before a discharge.

PHASE 3

Electrostatic Force Demonstration

Entries 18–23

Guiding Question

Can the electric field visibly move matter even though the machine's continuous current is very small?

WIMSHURST ELECTROSTATIC GENERATOR

Investigatory Project Report and Development Logbook

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To investigate the operation of the Wimshurst electrostatic generator, analyse the production of high voltage and low current, understand the role of capacitance using Leyden jars, measure voltage by spark gap, evaluate power output, detect electromagnetic emissions of the spark, and explore practical applications including the Tesla electrostatic power system.

LEGEND
Photo Placeholder - replace with actual experimental photograph
Observation / Data Recording
Question / Hypothesis
Conclusion / Finding

INVESTIGATORY PROJECT DEVELOPMENT LOGBOOK – 60 ENTRIES

PHASE 1
Conceptualization & Background Research
(Entries 1–7)

1. Selected the Wimshurst machine as the subject of the project.
2. Researched James Wimshurst (1832–1902) and electrostatic machines in the late 1900s.
3. Identified main investigation goals: high voltage, charge accumulation, capacitance, voltage measurement, power, and practical applications.
4. Defined key research questions and scope.
5. Reviewed basic safety requirements for high-voltage electrostatic equipment.
6. Searched available Wimshurst machines and selected the one used in experiments.
7. Studied influence machine designs: Wimshurst, Holtz, Pöppel (rotated Pöppel used variable capacitance via sector wing).

PHASE 2
Electrostatic Theory & Fundamentals
(Entries 8–15)

8. Reviewed electric charge: positive, negative, conductors, insulators, and charge conservation.
9. Studied surface charge density ($\sigma = Q/A$) and its effect on electric field strength.
10. Reviewed electric field (E) and its relationship to voltage: $E = V/d$ for parallel plates.
11. Reviewed capacitance (C) and the core relationship $Q = C \cdot V$.
12. Learned about Leyden jars (capacitors) storing charge and increasing energy.
13. Derived stored energy in a capacitor: $E = \frac{1}{2} C V^2$.
14. Studied air breakdown and dielectric strength of air ($\sim 3 \text{ kV/cm}$ under normal conditions).
15. Read about spark phenomena and the conditions required for electrical breakdown in air.

PHASE 3
Electrostatic Force Demonstration – Styrofoam Ball Attraction & Repulsion
(Entries 16–22)

16. Set up a lightweight Styrofoam ball suspended by a thin nylon thread between the Wimshurst terminals.
17. Observation: Neutral ball is attracted to a charged terminal (polarization).
18. Ball touches terminal $\rightarrow$ acquires same charge (conduction).
19. Observation: Ball is repelled from that same terminal.
20. Ball moves toward opposite terminal $\rightarrow$ attracted.
21. Ball touches opposite terminal $\rightarrow$ charge changes $\rightarrow$ repelled again.
22. Conclusion: Demonstrates attraction, charge transfer, and repulsion; electrostatic force does not require high current.

PHASE 4
Spark, Breakdown & Spark-Gap Measurement
(Entries 23–30)

23. Test the machine without Leyden jars, observed that blue.
24. Observed bright, thick, energetic spark with Leyden jars.
25. Decided to measure maximum repeatable spark gap (d).
26. Prepared a spark gap scale (see $V = E \cdot d$).
27. Increased gap until spark just failed; recorded distance.
28. Repeated multiple times for consistency.
29. Calculated estimated voltage from $V = E_{\text{break}} \cdot d$.
30. Compared results with and without Leyden jars.

PHASE 5
Leyden Jars – Capacitance & Energy
(Entries 31–38)

31. Connected Leyden jars to the machine.
32. Observed much thicker, brighter, louder sparks.
33. Measured maximum repeatable gap with jars.
34. Compared gap (inches) with and without jars.
35. Observed discharge frequency: slower with jars.
36. Observed brightness/energy: much greater with jars.
37. Explained: jars increase capacitance $\rightarrow$ store more charge.
38. Conclusion: Leyden jars store energy and deliver it in powerful bursts.

PHASE 6
Voltage, Current & Power
(Entries 39–45)

39. Estimated voltage from measured maximum gap.
40. Measured current with a sensitive microammeter.
41. Observed: current is very small (microamperes range).
42. Calculated instantaneous power: $P = V \cdot I$.
43. Observed despite high voltage, power is low.
44. Conclusion: Wimshurst produces high voltage, low current, low power.
45. Explained: energy comes in short bursts, not continuous.

PHASE 7
Electromagnetic Emission & AM Radio Detection
(Entries 46–53)

46. Connected battery-powered AM radio near the machine.
47. Tuned radio to an unused frequency.
48. Observed static or buzzing sounds from spark.
49. Moved antenna orientation and distance for clarity.
50. Compared sound intensity with and without Leyden jars.
51. Conclusion: spark emits electromagnetic waves.
52. Demonstrated relationship between electricity and magnetism (EM radiation).
53. Recorded best conditions for clear detection.

PHASE 8
Applications – Tesla Electrostatic Power System
(Entries 54–60)

54. Researched Nikola Tesla's T.T. (Coffin) and Tesla Coil.
55. Studied electrostatic power supply using Wimshurst machines.
56. Understood storage (Leyden jar) $\rightarrow$ converter $\rightarrow$ radio power.
57. Learned about electrostatic transformers and converters.
58. Compared electrostatic power with modern power systems.
59. Considered practical applications and historical significance.
60. Documented final findings, conclusions, and future work.

Charge Density & Electric Field
Higher charge density (σ) on electrode surfaces $\rightarrow$ stronger electric field (E) $\rightarrow$ when E exceeds the breakdown strength $\rightarrow$ spark (air becomes conductive).
 $E = V/d$
 $E_{\text{break}} \approx 3 \text{ kV/mm}$
 $V_{\text{break}} \approx E_{\text{break}} \times d$

Air Breakdown (Spark Condition)
Breakdown occurs when the electric field reaches the dielectric strength of air.

Capacitance & Energy
 $Q = C \cdot V$
 $E = \frac{1}{2} C V^2$
Leyden jars increase $C \rightarrow$ store more charge (Q) $\rightarrow$ store more energy (E).

Styrofoam Ball Interaction Sequence
1. Neutral ball approaches positive terminal.
2. Ball touches positive terminal and acquires positive charge.
3. Ball is repelled from positive terminal.
4. Ball is attracted toward negative terminal, touches and may change charge.
5. Ball is repelled from negative terminal.

Measuring the Spark Gap
The maximum gap at which the spark will just jump consistently is recorded. Distance d is used to calculate the voltage.

Effect of Leyden Jars
With Leyden Jars Connected
• Slower discharges (less frequent)
• Thin, bright, pointed sparks
• Greater stored energy per discharge
• Larger maximum spark gap (higher voltage)
Without Leyden Jars
• Rapid discharges (less frequent)
• Thin, faint, pointed sparks
• Much less energy per discharge
• Smaller maximum spark gap (lower voltage)

Voltage Estimate Example
 $d = 12 \text{ cm (0.012 m)}$
 $V = E_{\text{break}} \cdot d$
 $V = (3,000 \text{ V/mm}) \cdot (12 \text{ mm})$
 $V = 36,000 \text{ V (36 kV)}$

Typical Current Measurement
 $I = 5 - 20 \text{ }\mu\text{A (microamperes)}$
(Average value)
 $I \approx 0.000005 - 0.000020 \text{ A}$

Power Estimate Example
 $P = V \cdot I$
 $P = 36,000 \text{ V} \times 0.000010 \text{ A}$
 $P \approx 0.36 \text{ W (instantaneous)}$
Low power!

AM Radio Detection Setup (Battery-Powered Radio)
Place the radio near the machine for clear detection of the static noise from the spark on an unused AM frequency.

Radio Signal Comparison
Without Leyden Jars
• Faint, small static
• Lower amplitude
• Fainter signal
With Leyden Jars
• Slower, louder bursts
• Higher amplitude
• Stronger signal

Tesla Electrostatic Power Plant (1900s)
Historical photograph - replace if desired.

Electrostatic Power: From Generation to Useful Power
Wimshurst machine generates high voltage, low current. This is converted to a higher voltage, even higher power by a transformer. The resulting high voltage is used to power a radio receiver.

What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

Styrofoam Ball: Attraction → Charge Transfer → Repulsion

A lightweight suspended ball makes electrostatic force visible.

1

Neutral ball is attracted

Polarization makes the nearer side effectively opposite in charge.

2

Ball touches terminal

Charge transfers by conduction.

3

Ball is repelled

After contact it has the same polarity as that terminal.

4

Ball moves to opposite terminal

It is attracted, can touch, change charge, and be repelled again.

$$F = k |q_1 q_2| / r^2 \quad k \approx 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

What the Styrofoam Experiment Demonstrates

Electrostatic force depends on charge and separation—not on a large continuous current.

Why a neutral ball is attracted

The charged terminal polarizes the neutral ball. Opposite charge is effectively closer to the terminal than like charge, producing a net attractive force.

What changes after contact

Touch transfers charge. The ball can acquire the same polarity as the terminal and is then repelled.

Why this matters

The Wimshurst machine can produce a strong enough electric field to move a very light object even though its available continuous current is extremely small.

Unlike charges attract • Like charges repel

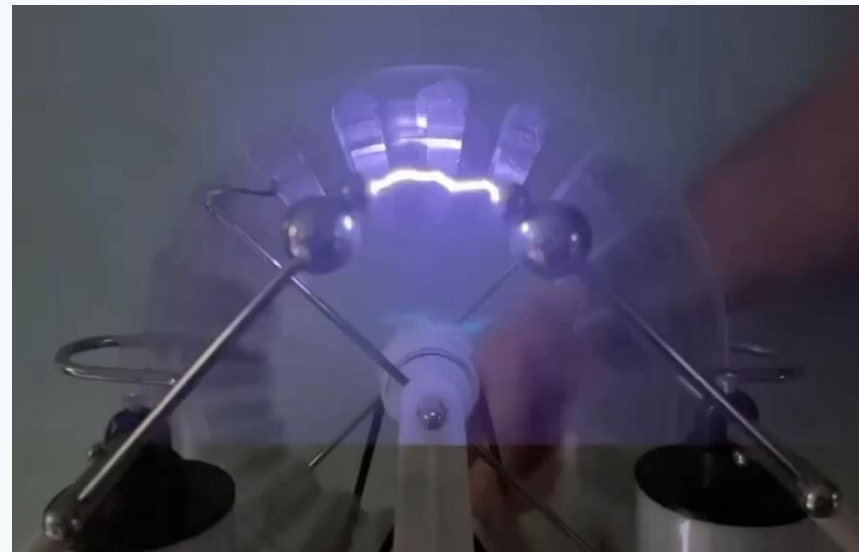
PHASE 4

Experimental Planning & Measurement Design

Entries 24–27

Guiding Question

How can we compare the two discharge conditions fairly and estimate the high voltage safely?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

Plan the Spark-Gap Experiment

Use the maximum repeatable spark gap as the primary quantitative measurement.

Measure

Use a ruler or caliper to measure electrode spacing. Repeat the measurement rather than relying on one spark.

Control

Keep electrode geometry, approximate cranking rate and room conditions as consistent as practical. Compare jars connected vs. disconnected.

Record

Gap distance • discharge frequency • brightness • thickness • sound • room conditions • Leyden-jar condition.

Voltage estimate: $V \approx (3 \text{ kV/mm}) \times d$

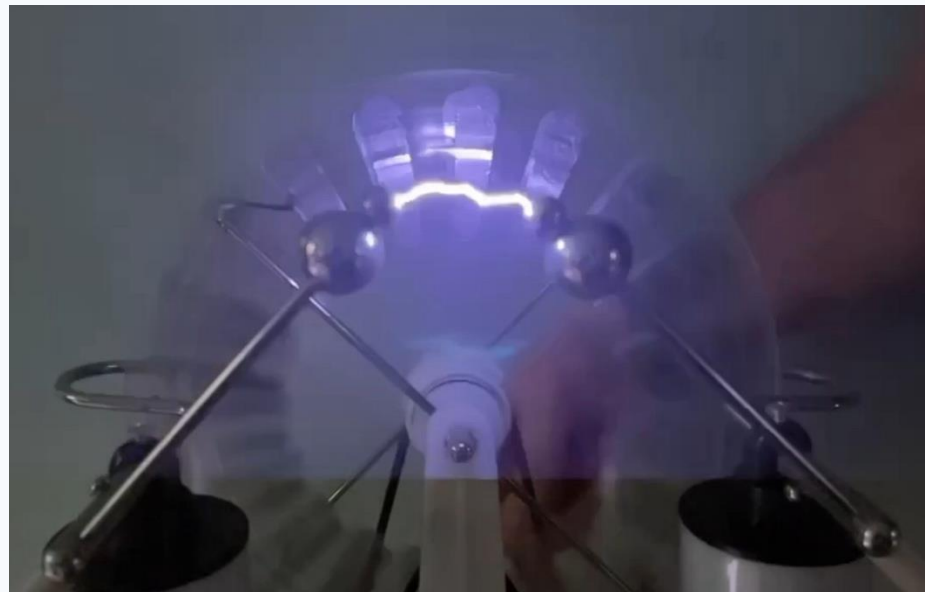
PHASE 5

Spark & Leyden-Jar Investigation

Entries 28–36

Guiding Question

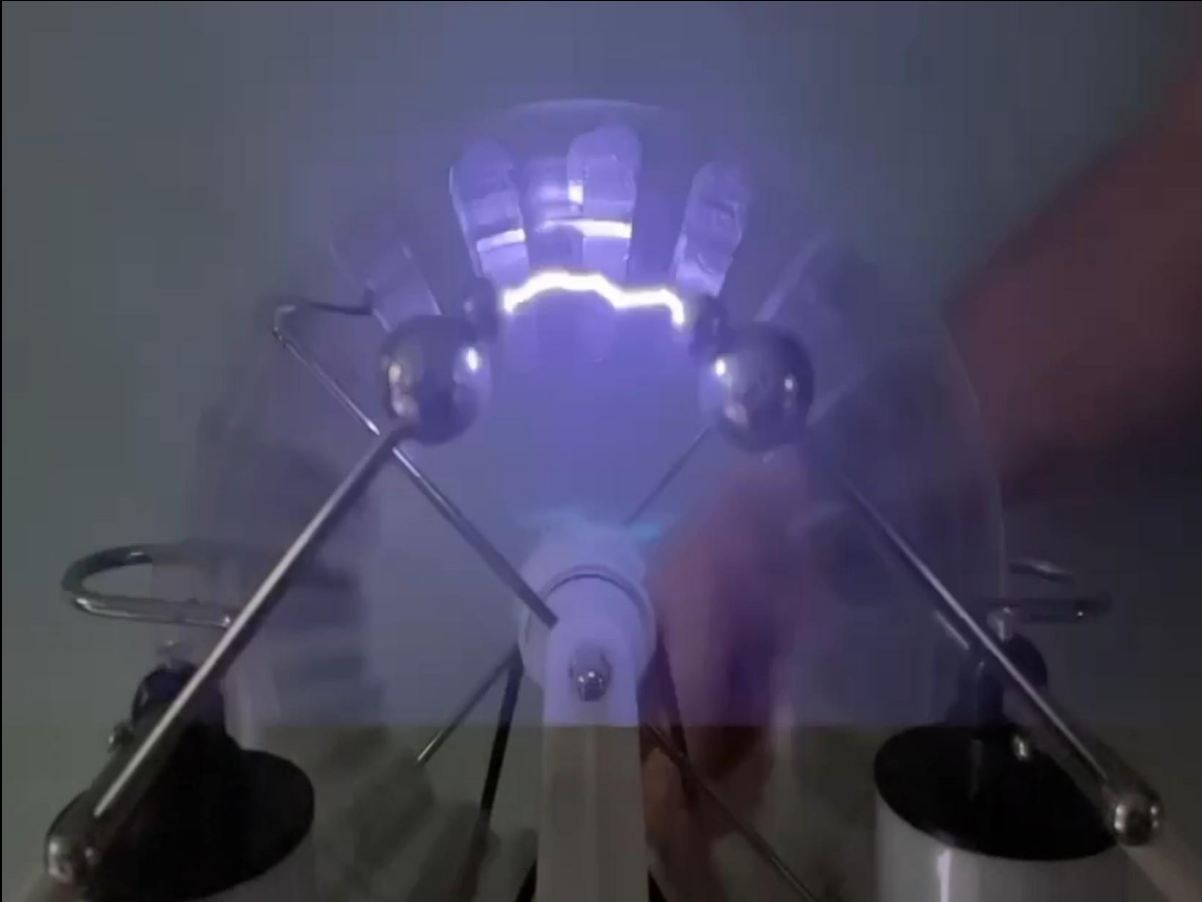
Why does removing the Leyden jars create a rapid thin blue discharge instead of large bright sparks?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

With Leyden Jars: Slower, Brighter, More Energetic Sparks



What changes?

- The jars greatly increase capacitance.
- More charge is needed to reach the breakdown region.
- Charging takes longer.
- Much more charge and energy are available when breakdown occurs.
- The result is a thick, bright, loud spark.

$$E = \frac{1}{2} C V^2$$

Without Leyden Jars: Rapid Thin Blue Discharge

Lower effective capacitance changes the discharge regime.

What is happening?

- Effective capacitance is small.
- A little transferred charge produces a large voltage rise.
- Surface charge creates a strong electric field.
- Air ionizes quickly.
- The small stored charge discharges, then the cycle repeats rapidly.

Key observation

The discharge can look like a rapid, thin blue stream rather than isolated bright sparks. Each event contains much less stored energy, but the machine reaches breakdown again very quickly.

Small $C \rightarrow$ little Q needed $\rightarrow$ breakdown reached quickly

Same Generator — Very Different Discharge

The Leyden jars primarily change charge storage and discharge energy.

WITHOUT LEYDEN JARS

- Low capacitance
- Voltage rises quickly
- Frequent discharge
- Thin blue stream
- Little energy per event

WITH LEYDEN JARS

- Higher capacitance
- More charge must accumulate
- Less frequent discharge
- Thick, bright spark
- Much more energy per event

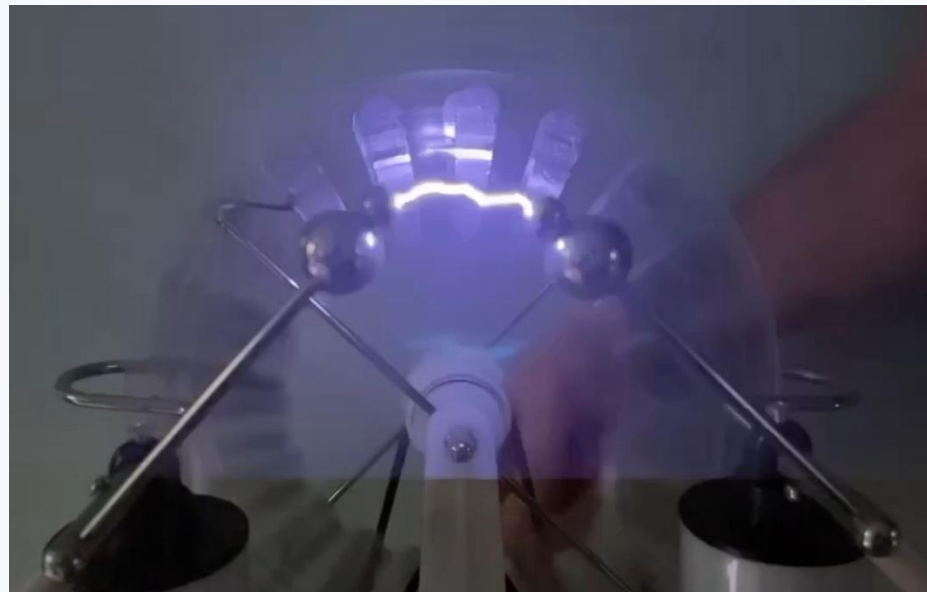
PHASE 6

Voltage, Current & Power Analysis

Entries 37–44

Guiding Question

How high is the voltage, why not measure it with an ordinary meter, and how much power does high voltage actually represent?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

Estimating Voltage with the Spark Gap

Air breakdown provides a practical indirect estimate.

$$V \approx (3 \text{ kV/mm}) \times d$$

Example

If the maximum repeatable gap is 10 mm:

$$10 \text{ mm} \times 3 \text{ kV/mm} \approx 30 \text{ kV}$$

Why it is only an estimate

Air breakdown varies with humidity, pressure, temperature, electrode shape, surface condition and other factors. Use several trials and report the maximum repeatable gap.

Why Not Use an Ordinary Voltmeter or Oscilloscope?

Direct measurement requires specialized high-voltage, very-high-impedance instrumentation.

Voltage rating

A Wimshurst machine may reach tens of kilovolts. Ordinary meters and probes are generally not rated for direct connection at these voltages.

Loading

A normal meter has finite input resistance. On a very-low-current electrostatic source, the meter can draw enough current to collapse the voltage instead of simply measuring it.

Oscilloscope ground

Many oscilloscopes use an earth-referenced probe ground. Connecting that ground to a floating electrostatic generator can alter the circuit and create an unintended discharge path.

High Voltage Does Not Necessarily Mean High Power

Power depends on both voltage and current.

$$P = V I$$

Illustrative Wimshurst calculation

$$V = 30,000 \text{ V}$$

$$I = 10 \text{ } \mu\text{A} = 0.000010 \text{ A}$$

$$P = (30,000)(0.000010) = 0.30 \text{ W}$$

Compare: USB supply

$$5 \text{ V} \times 2 \text{ A} = 10 \text{ W}$$

The USB supply has 6,000× less voltage yet can deliver more than 33× as much continuous power in this example.

30,000 volts sounds enormous — but at 10 millionths of an ampere, the calculated continuous power is only 0.30 watt.

Why Can a Low-Power Machine Make Such a Dramatic Spark?

Stored energy can be released in a very short time.

$$E = \frac{1}{2} C V^2$$

Energy accumulation

A Leyden jar may charge over a relatively long interval, then release much of its stored energy in a tiny fraction of a second.

$$P_{\text{discharge}} \approx E / t$$

Example only

$0.05 \text{ J} \div 0.0001 \text{ s} = 500 \text{ W}$ during that brief interval.

That does NOT mean the generator continuously produces 500 W.

Supplementary Demonstration: Lighting a Neon Lamp

High voltage can ionize neon and produce visible light with very small current.

What it demonstrates

A small neon lamp can glow or flash because the Wimshurst voltage is high enough to ionize the gas inside the lamp. This provides a second visible indication of high voltage besides the spark.

What it does NOT demonstrate

A glowing neon lamp does not imply large current or large continuous power. It reinforces the distinction between voltage, current and power.

High voltage $\neq$ high current $\neq$ high power

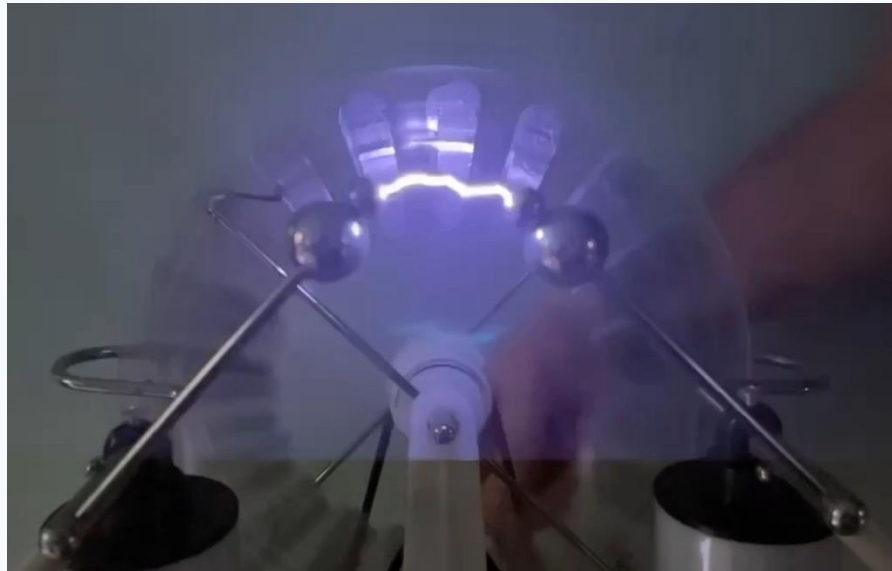
PHASE 7

Electromagnetic Emission Investigation

Entries 45–50

Guiding Question

Does the spark also send out an electromagnetic signal that can be detected without a wire connection?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

A Spark Is Also an Electromagnetic Signal

A rapid electrical discharge produces a broadband electromagnetic disturbance.

Spark → fast-changing current/field → electromagnetic pulse → AM radio click

Historical connection

Early spark-gap transmitters intentionally used electrical sparks to generate radio-frequency energy.

Why compare the jars?

The two discharge regimes differ in repetition rate and energy per event, so the radio should provide an audible comparison.

Experiment: Detect the Spark with an AM Radio

Compare the radio interference with and without the Leyden jars.

Setup

1. Battery-powered AM radio
2. Tune between stations
3. Start about 1–2 m away
4. Keep volume fixed
5. Use the same spark gap and similar cranking speed
6. Optionally compare maximum clear detection distance

Without Leyden Jars

Expected: rapid, weaker impulses may sound like fast crackling or buzzing.

The discharge repeats quickly because little charge is stored per event.

With Leyden Jars

Expected: less frequent but stronger individual clicks or pops.

Each spark releases more stored energy.

PHASE 8

Practical Applications & Testatika Extension

Entries 51–55

Guiding Question

How can electrostatic high voltage be useful, and what additional stages would be needed to supply conventional loads?



What this phase adds

Research and observations are organized so each phase builds the explanation used in the final demonstration.

Practical Uses of High Electrostatic Voltage

High voltage itself is useful even when available current is small.

Electrostatic precipitation

Remove particles from air or exhaust streams.

Electrostatic coating

Attract charged paint or powder to a target surface.

Ionization

Create ions for demonstrations, sensing or processing.

Particle acceleration

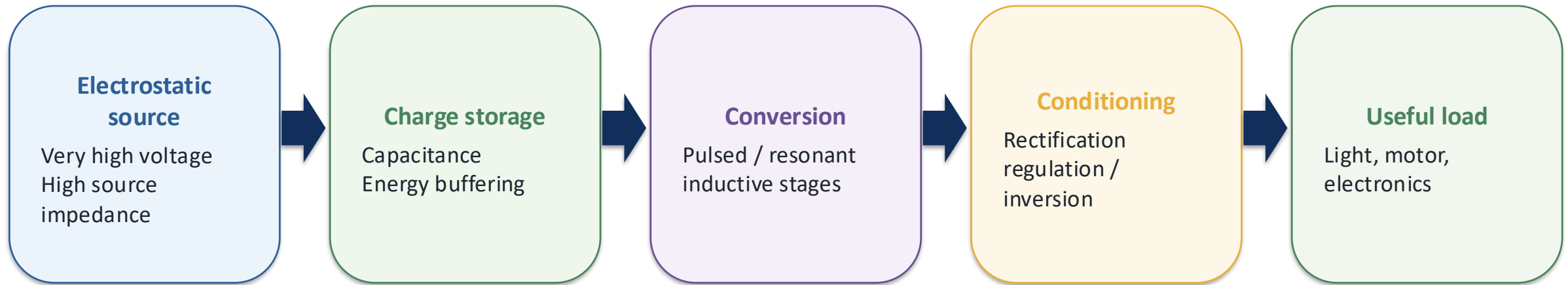
Use electric fields to accelerate charged particles.

High-voltage research

Study insulation, corona, breakdown and electric fields.

From Electrostatic Generation to Practical Electrical Power

The central engineering problem is not just producing voltage—it is converting and conditioning energy for a load.

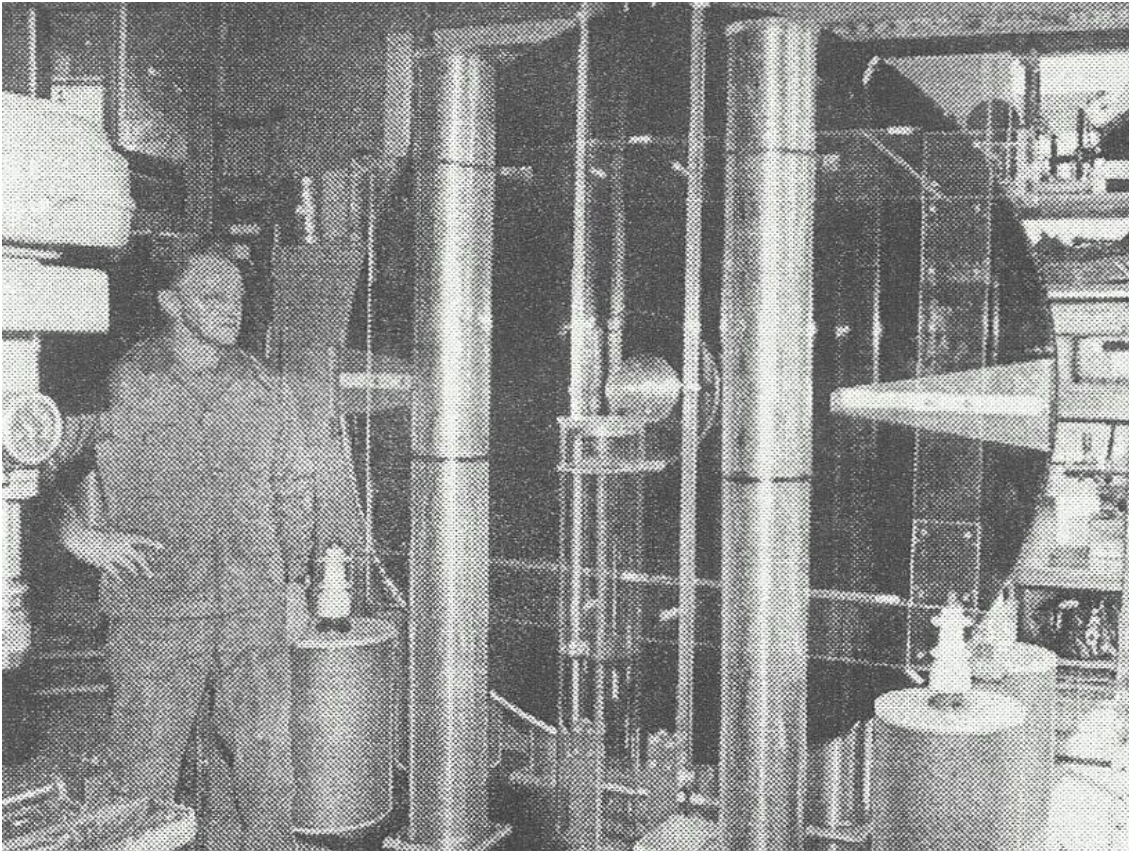


Inductive loads

Motors and transformers may require starting current, reactive current and transient handling. A practical conversion system must isolate the electrostatic source from those load demands.

Testatika: A Reported Extension of the Influence-Machine Idea

For this project, it is presented under the premise that its reported operation is valid.



Conceptual relationship

The electrostatic disk system can be viewed as a high-voltage front end, followed by additional storage, inductive/resonant conversion and output-conditioning structures.

Important distinction

Photographs establish that the Testatika contains much more than rotating disks. Exact internal wiring and the function of every component are not fully documented in public primary engineering drawings, so detailed circuit descriptions should be labeled as reported or reconstructed.

Final Phase — Analysis, Documentation & Presentation

Entries 56–60: assemble the observations into one coherent scientific explanation.

Final Demonstration Sequence

1. Styrofoam attraction / repulsion
2. Spark gap
3. Compare jars vs. no jars
4. Estimate voltage
5. Explain voltage, current & power
6. Neon lamp demonstration
7. AM-radio detection
8. Practical applications / Testatika

What the project establishes

- Electrostatic force can act without large current.
- Capacitance changes charge and energy storage.
- Air breakdown provides an indirect voltage estimate.
- High voltage can coexist with low continuous power.
- Sparks produce detectable electromagnetic emissions.

What to record

Replace AI placeholders with actual photos where possible. Record measured spark gaps, room conditions, qualitative discharge observations and AM-radio results. Distinguish measurements from illustrative examples.

Overall Conclusion

The coherent story

Mechanical motion separates charge. Charge creates electric fields and electrostatic force. Capacitance determines how much charge and energy can accumulate. Air breaks down when the electric field becomes strong enough. The resulting spark can be used to estimate voltage and can radiate a detectable electromagnetic pulse. Yet high voltage alone does not imply high continuous power. Practical power delivery requires storage, conversion, conditioning and load matching.

Electrostatic force → high voltage → capacitance → breakdown → power → electromagnetism → applications

Safety

High voltage requires respect.

- Do not deliberately discharge the machine through a person.
- Treat Leyden jars as charged capacitors even after cranking stops.
- Discharge them with an appropriate insulated discharge tool before touching conductors or changing connections.
- Keep away from flammable vapors and sensitive electronics.
- Do not connect ordinary meters or oscilloscope probes directly to the generator.
- Keep people with implanted electronic medical devices away from the discharge area.
- Perform the demonstration with responsible adult supervision.

Perspective

High-voltage sources capable of sustained current can be far more dangerous than a Wimshurst machine.

Project Source Materials

The deck reorganizes the existing presentation content according to the authoritative logbook flow.

Primary flow

Wimshurst_60_Entry_Logbook_A4_With_Styrofoam_Experiment.docx — phase sequence, entries, Styrofoam experiment, formulas and final demonstration progression.

Existing presentation content

Wimshurst_Electrostatic_Generator_Presentation_Expanded.pptx — voltage/current/power explanation, spark-gap measurement, meter/scope limitations, Leyden-jar comparison, neon lamp, AM-radio experiment, safety and Testatika extension.

Supporting Testatika material

Testatika_History_Architecture_and_Power_Conversion_Report — used only to preserve the distinction between visible/reported architecture and later reconstruction hypotheses.

Holtz_Wimshurst_Pidgeon_Testatika_Student_Research_Guide — supplemental research guide examining the evolution from Holtz to Wimshurst to Pidgeon and ultimately the Testatika, emphasizing electrostatic induction, variable capacitance, charge storage, and the development of electrostatic energy-conversion concepts.

Research Extension: Converting Wimshurst Output into Usable Power

Treat the machine as a high-voltage charge source feeding an energy-conversion chain.

Wimshurst → HV storage → controlled pulse → LC / transformer → rectification → output storage → load



The converter does not create energy.

It transforms a very-high-voltage/high-impedance source into a lower-voltage/higher-current form, with losses at each stage.

Could Approximately 0.25 W Be Achieved?

The answer depends on measured average charging power.

Illustrative source

$$\begin{aligned} &25,000 \text{ V} \times 20 \text{ } \mu\text{A} \\ &= 25,000 \times 0.000020 \\ &= 0.50 \text{ W} \end{aligned}$$

Illustrative conversion

$$\begin{aligned} &90\% \times 70\% \times 85\% \times 90\% \\ &\approx 48\% \\ &0.50 \text{ W} \times 0.48 \approx 0.24 \text{ W} \end{aligned}$$

These values are examples—not measurements of this Wimshurst machine.

Measure a known capacitance and charging time: $E = \frac{1}{2}CV^2$ and $P_{\text{avg}} \approx E/t$.
This determines whether the 0.25 W target is realistic for the actual apparatus.

Proof-of-Concept Development Sequence

Build and measure one energy-conversion stage at a time.

1. Characterize charging of a known, safely rated capacitance.
2. Calculate stored energy $E = \frac{1}{2}CV^2$ and average charging power E/t .
3. Transfer stored energy into an enclosed inductive / resonant stage.
4. Step down the pulse with a transformer or LC/transformer network.
5. Rectify and buffer the lower-voltage output.
6. Apply a calibrated resistive load and measure V_{out} and I_{out} .
7. Calculate $P_{out} = V_{out}I_{out}$ and conversion efficiency.
8. Demonstrate a small LED/load after the measurements are understood.

Safety: added high-voltage capacitance can store dangerous energy. Keep the HV section enclosed, intentionally discharged, and adult-supervised.